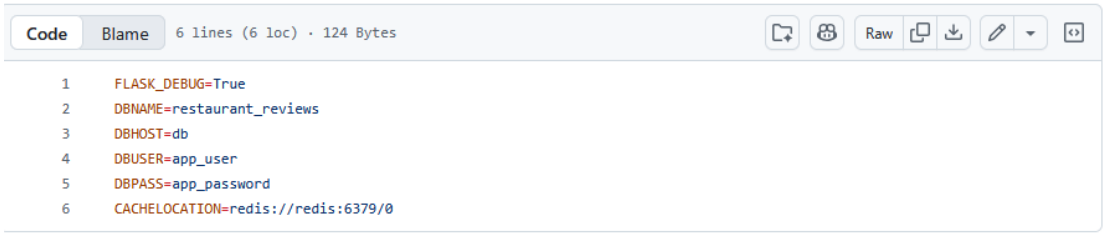
Homework DS - Enrichment Session: Exploring SQL & GitHub

Digital Skola Data Science Batch 66
RIYEN PERDANA

1. Login github.
2. Lakukan fork terhadap repository https://github.com/juggdante/msdocs-flask-postgresql-sample-app-azd.git, dengan cara menekan tombol *fork* pada sudut kanan atas, secara otomatis projek akan ada pada respository anda.
3. Database setup, untuk melakukan hal tersebut ikuti langkah langkah berikut :
   a. *Create a new database*;

      Proses pembuatan database dilakukan dengan cara melakukan proses edit data pada beberapa file, antara lain :

      - *.env*



      - *.devcontainer/docker-compose.yml*

```
db:
  image: postgres:15
  restart: unless-stopped
  volumes:
    - postgres-data:/var/lib/postgresql/data
  environment:
    POSTGRES_DB: restaurant_reviews
    POSTGRES_USER: app_user
    POSTGRES_PASSWORD: app_password
  healthcheck:
    test: ["CMD-SHELL", "pg_isready -U app_user -d restaurant_reviews -h localhost"]
    interval: 5s
    timeout: 5s
    retries: 5

  # Add "forwardPorts": ["5432"] to **devcontainer.json** to forward PostgreSQL locally.
  # (Adding the "ports" property to this file will not forward from a Codespace.)
```

- *.devcontainer/devcontainer.json. bagian koneksi SQLTools*

```
"sqltools.connections": [
    {
        "name": "Container database",
        "driver": "PostgreSQL",
        "previewLimit": 50,
        "server": "db",
        "port": 5432,
        "database": "restaurant_reviews",
        "username": "app_user",
        "password": "app_password"
    }
],
```

b. Create two tables;

Untuk proses create tables, disini langsung pembuatan table *menu* selain table *restaurant* dan *review*

```
from sqlalchemy import Column, DateTime, ForeignKey, Integer, Numeric, String, Text, UniqueConstraint
from sqlalchemy.orm import validates

from app import db


class Restaurant(db.Model):
    __tablename__ = 'restaurant'
    id = Column(Integer, primary_key=True)
    name = Column(String(50))
    street_address = Column(String(50))
    description = Column(String(250))

    def __str__(self):
        return self.name

class Review(db.Model):
    __tablename__ = 'review'
    id = Column(Integer, primary_key=True)
    restaurant_id = Column(Integer, ForeignKey('restaurant.id', ondelete="CASCADE"))
    user_name = Column(String(30))
    rating = Column(Integer)
    review_text = Column(String(500))
    review_date = Column(DateTime)

    @validates('rating')
    def validate_rating(self, key, value):
        assert value is None or (1 <= value <= 5)
        return value

    def __str__(self):
        return f"{self.user_name}: {self.review_date:%x}"

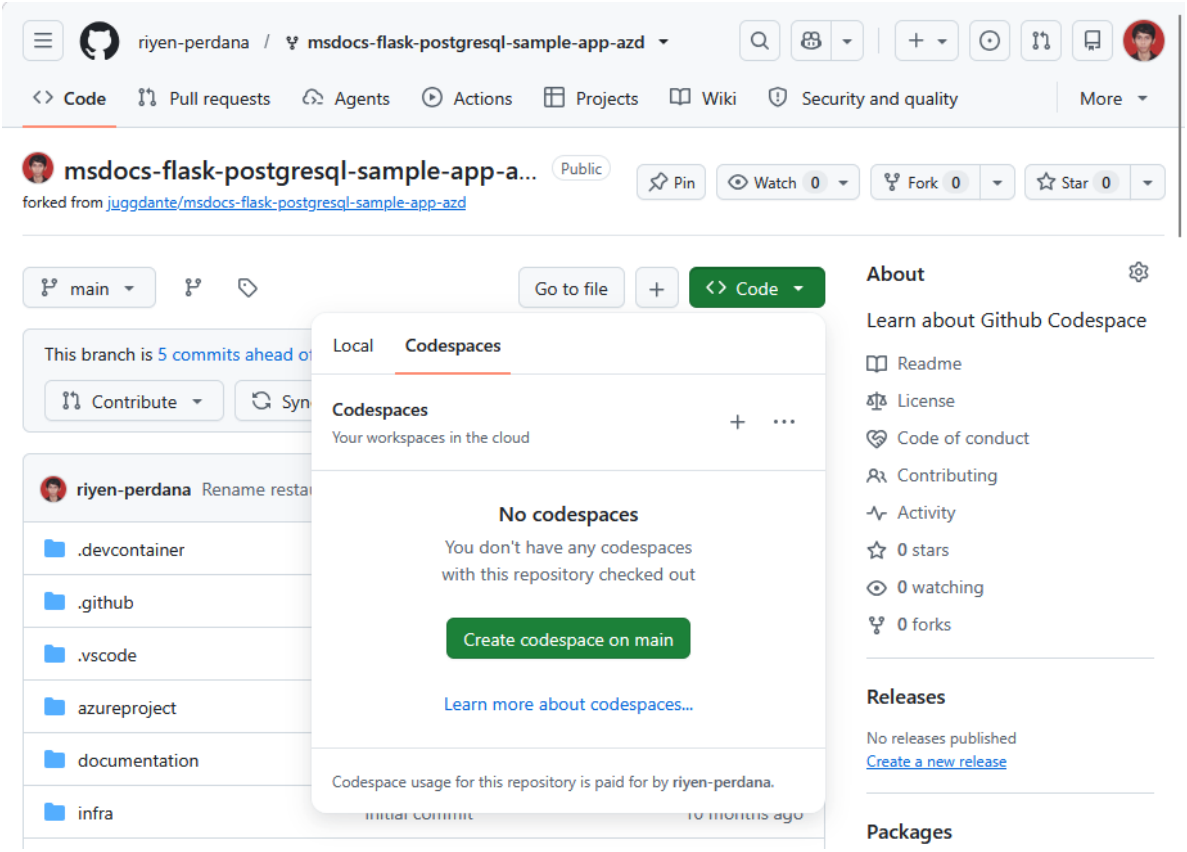
class Menu(db.Model):
    __tablename__ = 'menu'

    id = Column(Integer, primary_key=True)
    restaurant_id = Column(Integer, ForeignKey('restaurant.id', ondelete="CASCADE"), nullable=False)
    item_name = Column(String(100), nullable=False)
    price = Column(Numeric(10, 2), nullable=False)
    description = Column(Text)

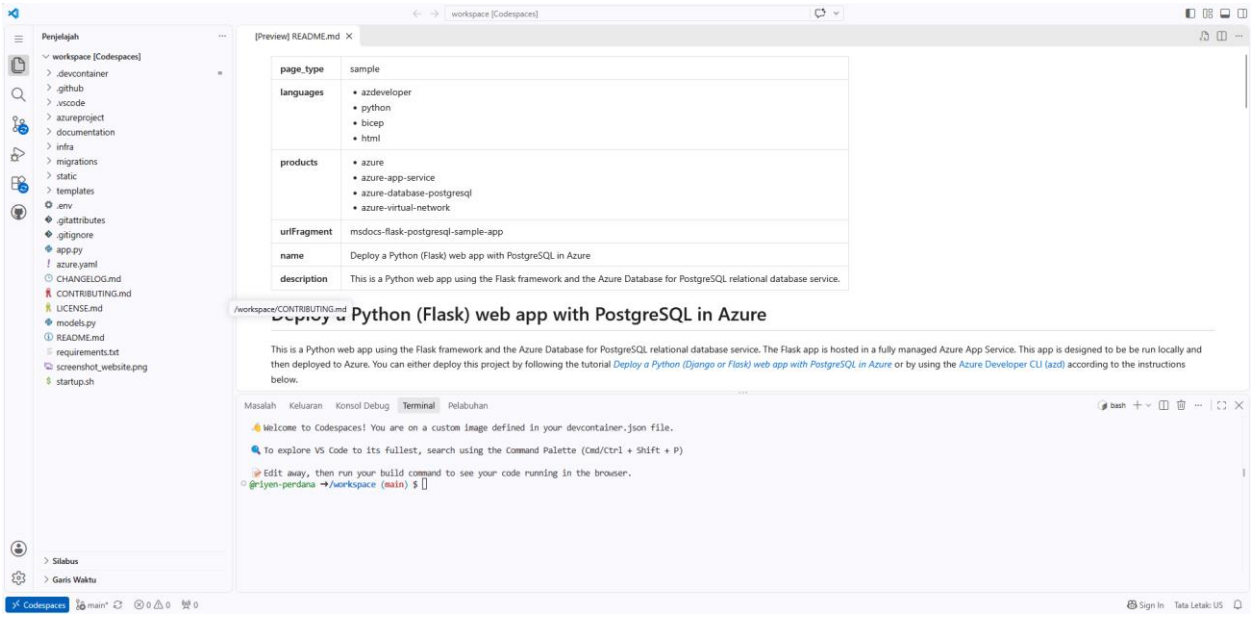
    # Menjaga agar nama menu tidak boleh sama dalam 1 restoran yang sama
    __table_args__ = (
        UniqueConstraint('restaurant_id', 'item_name', name='uq_restaurant_item'),
    )

    def __str__(self):
        return f"{self.item_name} - {self.price}"
```

4. Untuk mulai menggunakan codespaces, silahkan klik bagian code yang berwarna hijau, pilih tab codespaces dan klik Create codespace on main, ini akan membuka tampilan editor pada github.



5. Tampilan codebase projek repository



6. Untuk mulai menggunakan projek lakukan hal berikut pada terminal :

   a. python3 -m pip install -r requirements.txt

```
@riyen-perdana ➜/workspace (main) $ python3 -m pip install -r requirements.txt
Building wheels for collected packages: psycopg2
  Building wheel for psycopg2 (pyproject.toml) ... done
  Created wheel for psycopg2: filename=psycopg2-2.9.13-cp312-cp312-linux_x86_64.whl size=530321 sha256=464235d76e81d2c9dadea949578e4c72c3bc439555c5c947c868f6c70d121e96
  Stored in directory: /home/vscode/.cache/pip/wheels/62/5f/66/4a6327ff8c4001a27e5b76cbc09a614092449fde51fcbf3375
Successfully built psycopg2
Installing collected packages: typing-extensions, python-dotenv, psycopg2, markupsafe, itsdangerous, click, blinker, wtforms, werkzeug, SQLAlchemy, Mako, jinja2, Flask, alembic, flask_wtf, Flask-SQLAlchemy, Flask-
Migrate
Successfully installed Flask-3.1.3 Flask-Migrate-4.1.0 Flask-SQLAlchemy-3.1.1 Mako-1.4.3 SQLAlchemy-2.1.6 alembic-1.20.0 blinker-1.9.0 click-8.5.0 flask_wtf-1.3.0 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.4 p
sycopg2-2.9.13 python-dotenv-1.2.4 typing-extensions-4.16.0 werkzeug-3.1.9 wtforms-3.2.2

[notice] A new release of pip is available: 25.0.1 -> 26.2.2
[notice] To update, run: pip install --upgrade pip
```

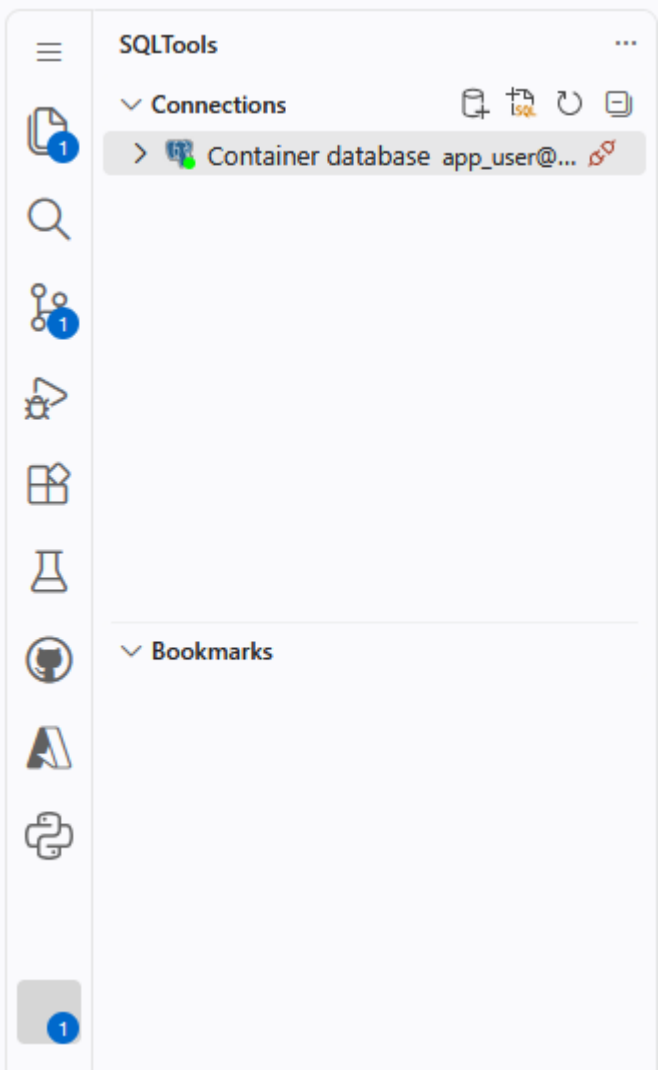
   b. python3 -m flask db upgrade

```
@riyen-perdana ➜/workspace (main) $ python3 -m flask db upgrade
Loading config.development and environment variables from .env file.
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
INFO  [alembic.runtime.migration] Running upgrade  -> d0c7b8e4b57c, Initial migration.
```

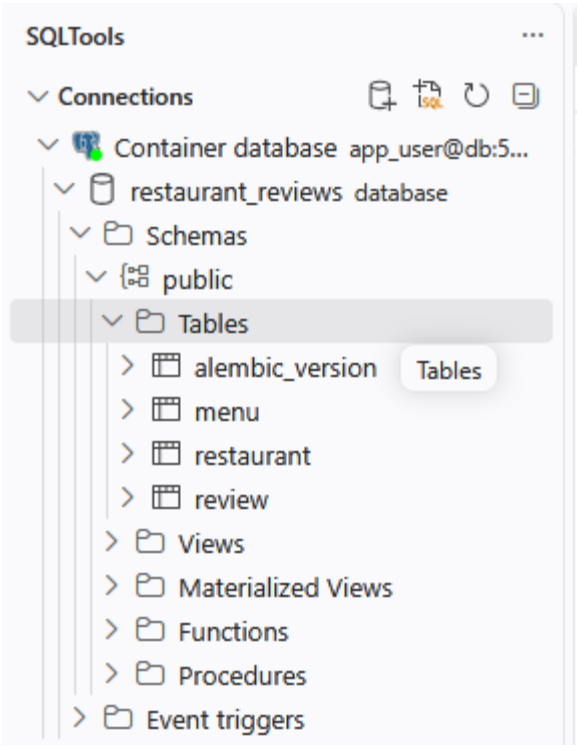
   c. python3 -m flask run

```
@riyen-perdana ➜/workspace (main) $ python3 -m flask run
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
Loading config.development and environment variables from .env file.
 * Debugger is active!
 * Debugger PIN: 100-039-673
Request for index page received
127.0.0.1 - - [21/Oct/2026 23:18:59] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [21/Oct/2026 23:18:59] "GET /static/images/azure-icon.svg HTTP/1.1" 200 -
127.0.0.1 - - [21/Oct/2026 23:19:00] "GET /static/favicon.ico HTTP/1.1" 200 -
```

7. Buka SQLTools dan koneksikan database

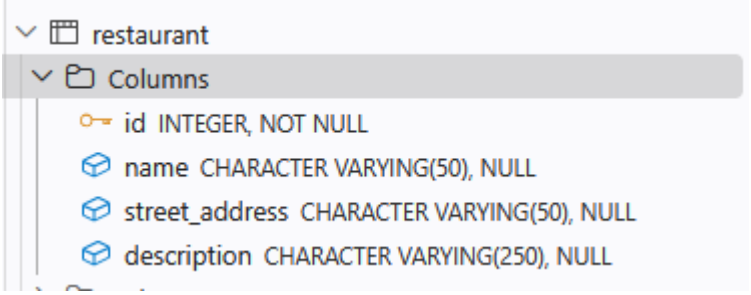


8. Proses migrasi database dan tabel dapat dilihat pada gambar di bawah ini

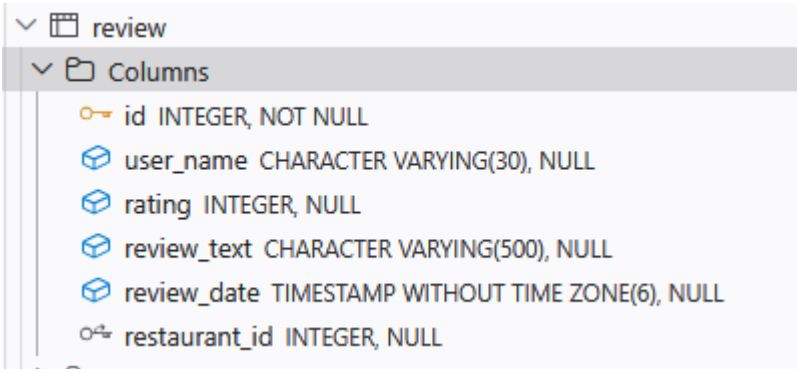


Untuk nama database sudah menjadi restaurant_reviews, diikuti dengan beberapa tabel menu, restaurant dan review.

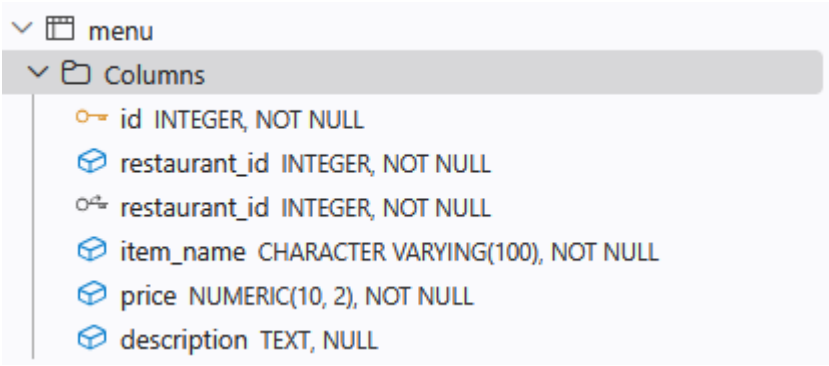
9. Schema table restaurant



10. Schema table review



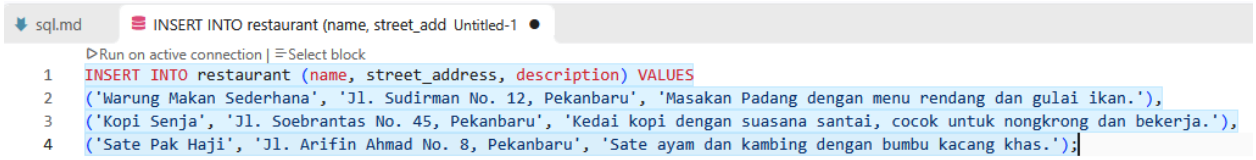
11. Schema table menu

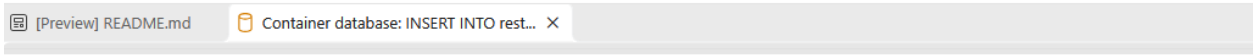


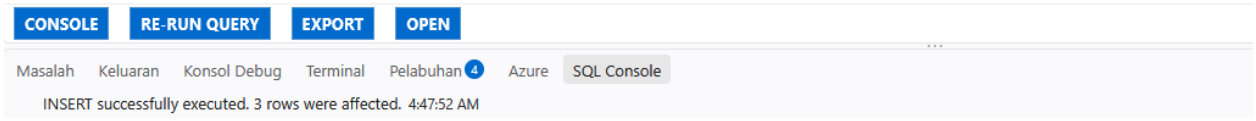
INSERTING DATA

12. Insert 3 (tiga) data restaurant

```
INSERT INTO restaurant (name, street_address, description) VALUES
('Warung Makan Sederhana', 'Jl. Sudirman No. 12, Pekanbaru', 'Masakan Padang dengan menu rendang dan gulai ikan.'),
('Kopi Senja', 'Jl. Soebrantas No. 45, Pekanbaru', 'Kedai kopi dengan suasana santai, cocok untuk nongkrong dan bekerja.'),
('Sate Pak Haji', 'Jl. Arifin Ahmad No. 8, Pekanbaru', 'Sate ayam dan kambing dengan bumbu kacang khas.');
```





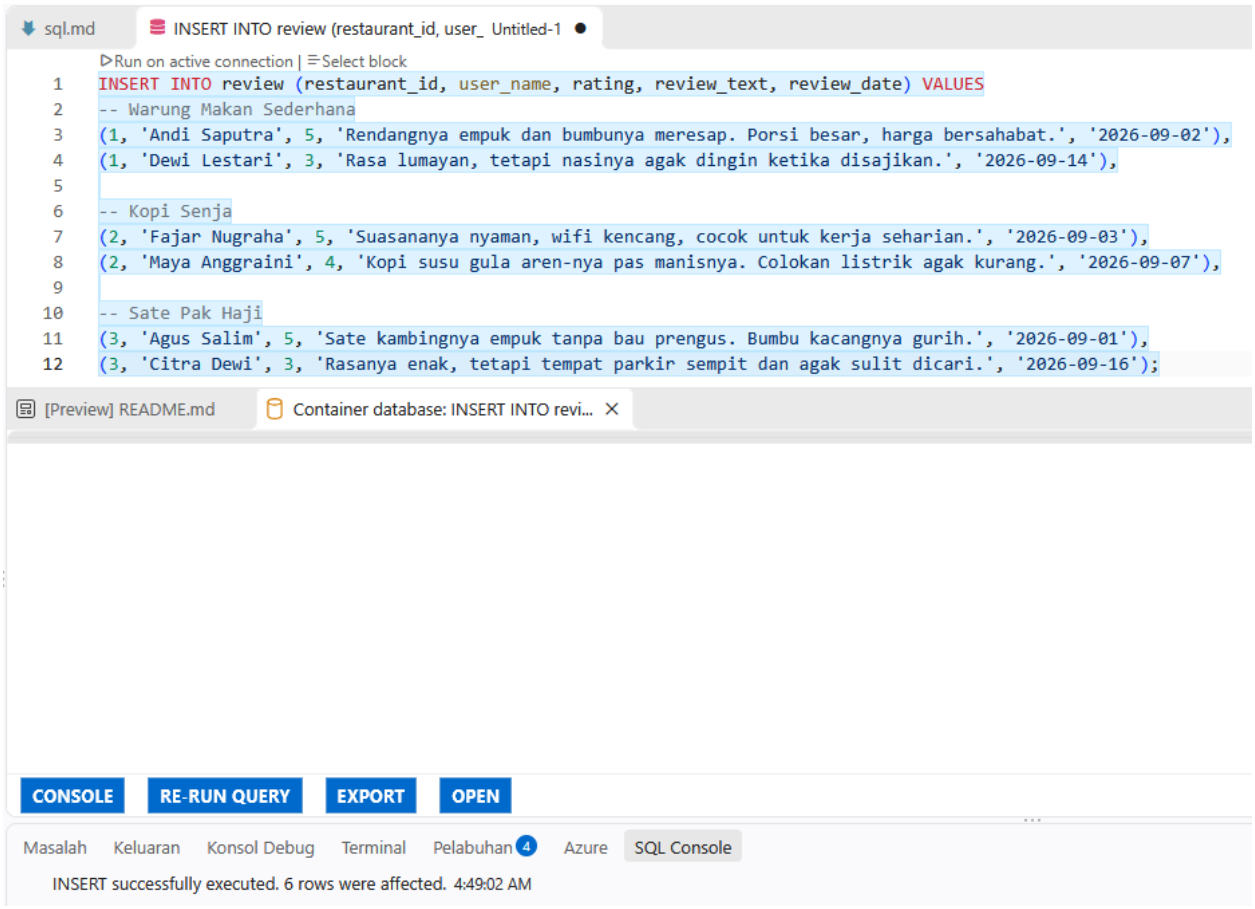


13. Insert data review

```
INSERT INTO review (restaurant_id, user_name, rating, review_text, review_date) VALUES
-- Warung Makan Sederhana
(1, 'Andi Saputra', 5, 'Rendangnya empuk dan bumbunya meresap. Porsi besar, harga bersahabat.', '2026-09-02'),
(1, 'Dewi Lestari', 3, 'Rasa lumayan, tetapi nasinya agak dingin ketika disajikan.', '2026-09-14'),

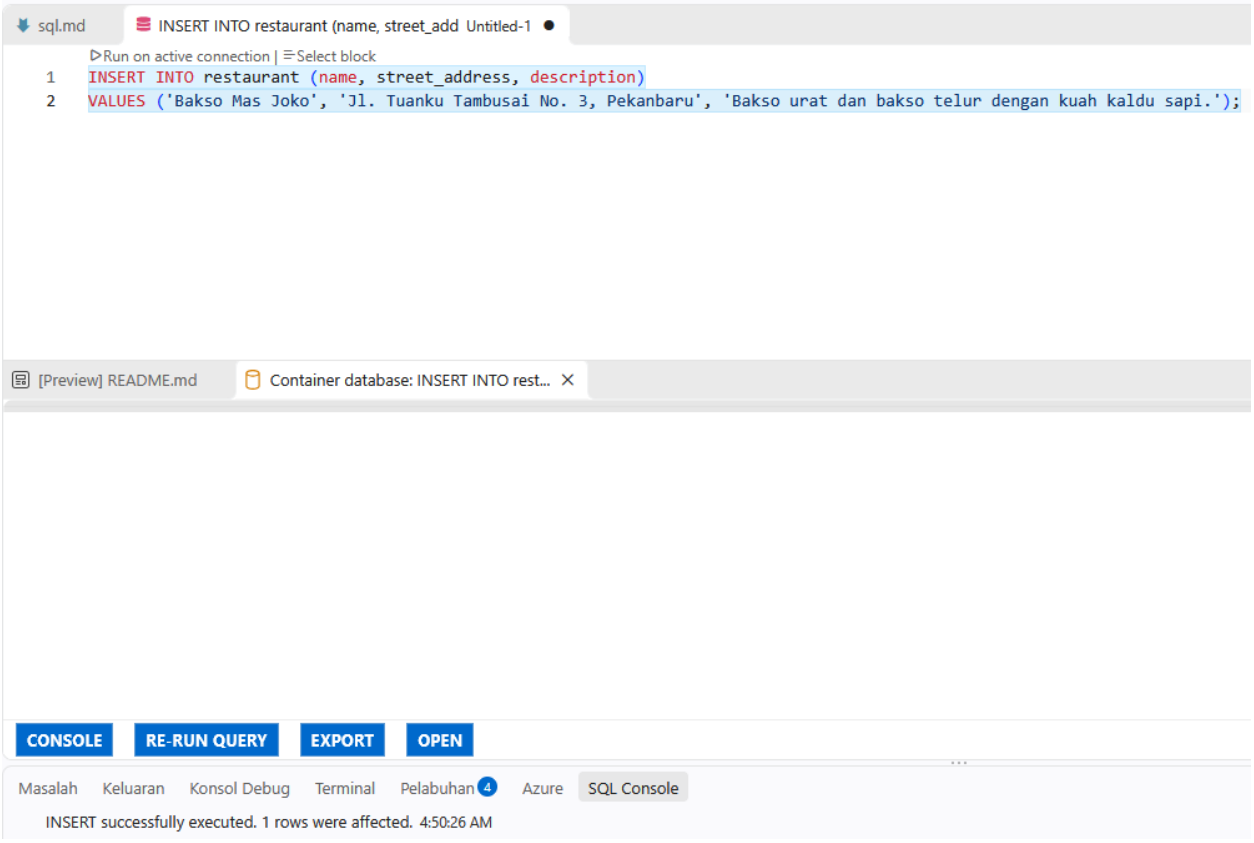
-- Kopi Senja
(2, 'Fajar Nugraha', 5, 'Suasananya nyaman, wifi kencang, cocok untuk kerja seharian.', '2026-09-03'),
(2, 'Maya Anggraini', 4, 'Kopi susu gula aren-nya pas manisnya. Colokan listrik agak kurang.', '2026-09-07'),

-- Sate Pak Haji
(3, 'Agus Salim', 5, 'Sate kambingnya empuk tanpa bau prengus. Bumbu kacangnya gurih.', '2026-09-01'),
(3, 'Citra Dewi', 3, 'Rasanya enak, tetapi tempat parkir sempit dan agak sulit dicari.', '2026-09-16');
```


sql.md
INSERT INTO review (restaurant_id, user_ Untitled-1
Run on active connection | Select block
INSERT INTO review (restaurant_id, user_name, rating, review_text, review_date) VALUES
-- Warung Makan Sederhana
(1, 'Andi Saputra', 5, 'Rendangnya empuk dan bumbunya meresap. Porsi besar, harga bersahabat.', '2026-09-02'),
(1, 'Dewi Lestari', 3, 'Rasa lumayan, tetapi nasinya agak dingin ketika disajikan.', '2026-09-14'),
-- Kopi Senja
(2, 'Fajar Nugraha', 5, 'Suasananya nyaman, wifi kencang, cocok untuk kerja seharian.', '2026-09-03'),
(2, 'Maya Anggraini', 4, 'Kopi susu gula aren-nya pas manisnya. Colokan listrik agak kurang.', '2026-09-07'),
-- Sate Pak Haji
(3, 'Agus Salim', 5, 'Sate kambingnya empuk tanpa bau prengus. Bumbu kacangnya gurih.', '2026-09-01'),
(3, 'Citra Dewi', 3, 'Rasanya enak, tetapi tempat parkir sempit dan agak sulit dicari.', '2026-09-16');
[Preview] README.md
Container database: INSERT INTO revi...
CONSOLE
RE-RUN QUERY
EXPORT
OPEN
Masalah
Keluaran
Konsol Debug
Terminal
Pelabuhan 4
Azure
SQL Console
INSERT successfully executed. 6 rows were affected. 4:49:02 AM

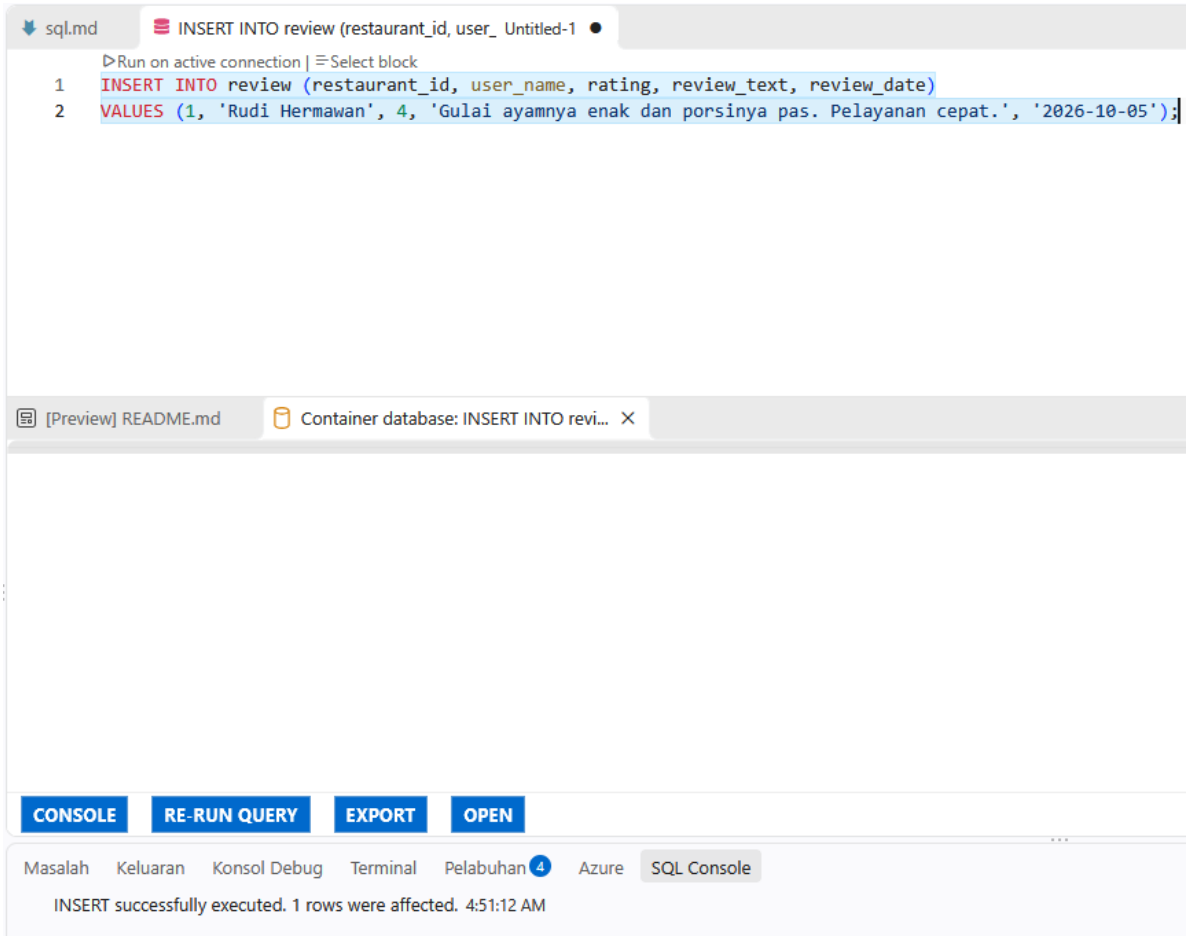
14. Insert kan 1 data restaurant baru

```
INSERT INTO restaurant (name, street_address, description)
VALUES ('Bakso Mas Joko', 'Jl. Tuanku Tambusai No. 3, Pekanbaru',
'Bakso urat dan bakso telur dengan kuah kaldu sapi.');
```



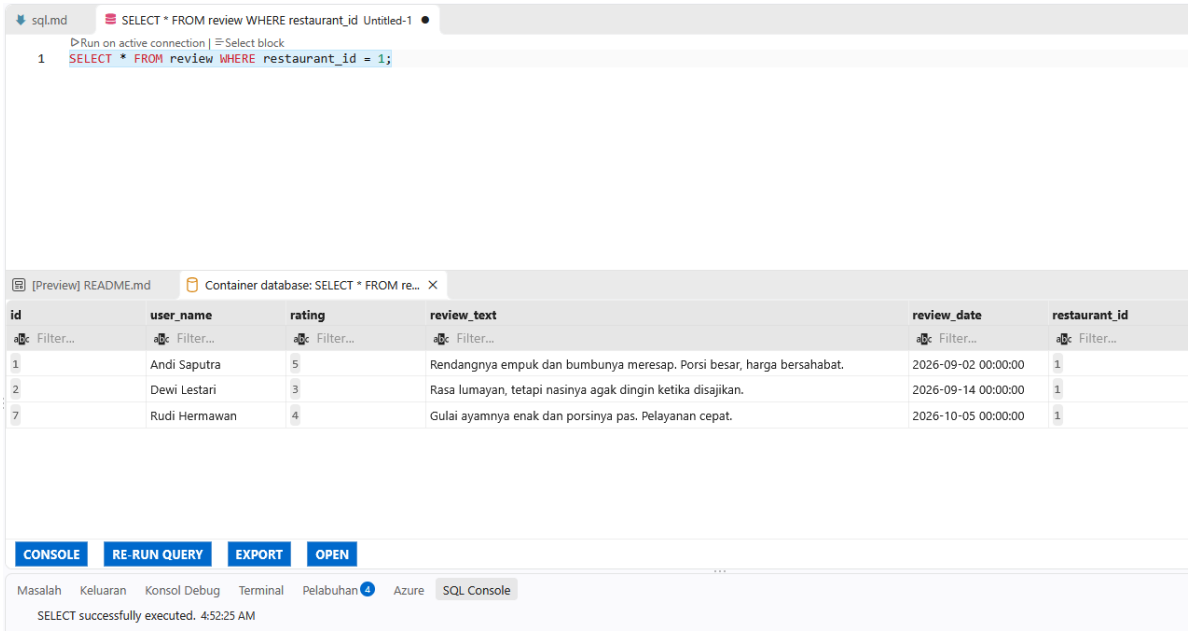
15. Insert kan 1 data review baru

```
INSERT INTO review (restaurant_id, user_name, rating, review_text, review_date)
VALUES (1, 'Rudi Hermawan', 4, 'Gulai ayamnya enak dan porsinya pas. Pelayanan cepat.', '2026-10-05');
```



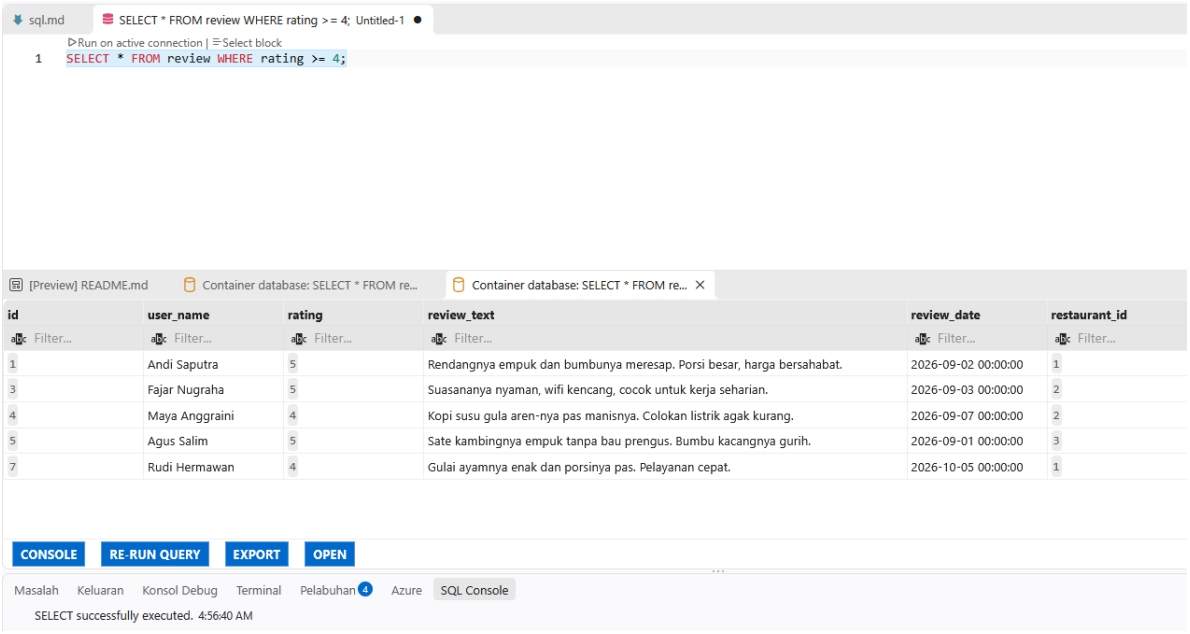
16. Select data review berdasarkan restaurant_id

```
SELECT * FROM review WHERE restaurant_id = 1;
```



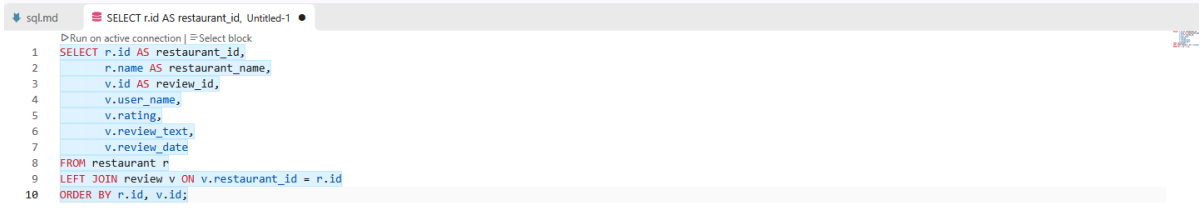
17. Select data review dengan rating lebih dari 4

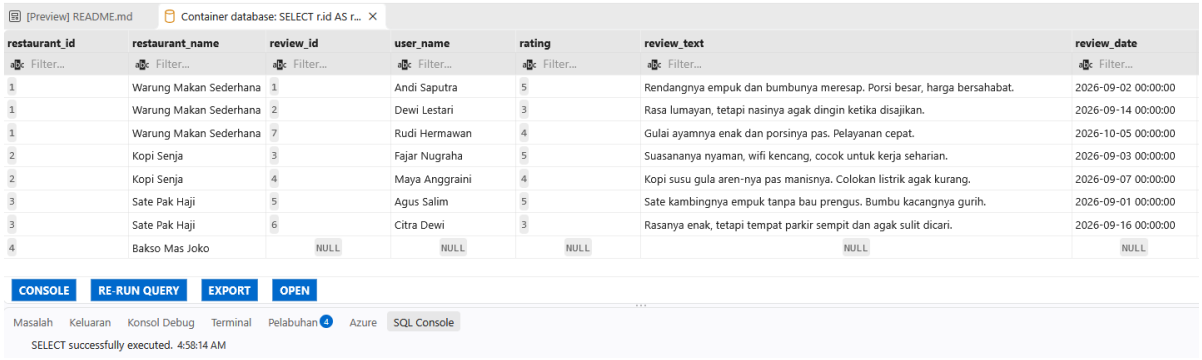
```
SELECT * FROM review WHERE rating >= 4;
```



18. Gunakan JOIN untuk menampilkan daftar restoran beserta review nya.

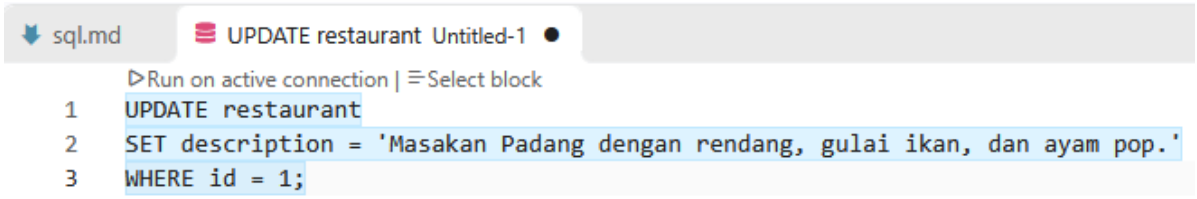
```
SELECT r.id AS restaurant_id,
       r.name AS restaurant_name,
       v.id AS review_id,
       v.user_name,
       v.rating,
       v.review_text,
       v.review_date
FROM restaurant r
LEFT JOIN review v ON v.restaurant_id = r.id
ORDER BY r.id, v.id;
```

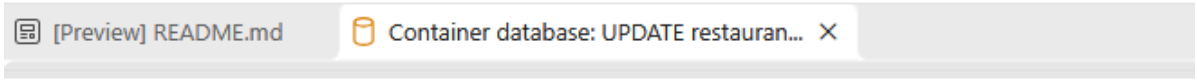


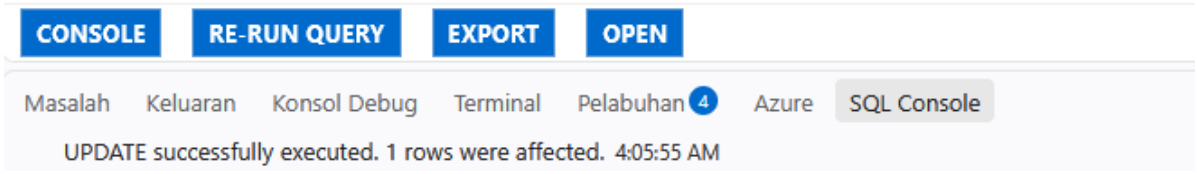


19. Update description salah satu restaurant

```
UPDATE restaurant
SET description = 'Masakan Padang dengan rendang, gulai ikan, dan ayam pop.' WHERE id = 1;
```

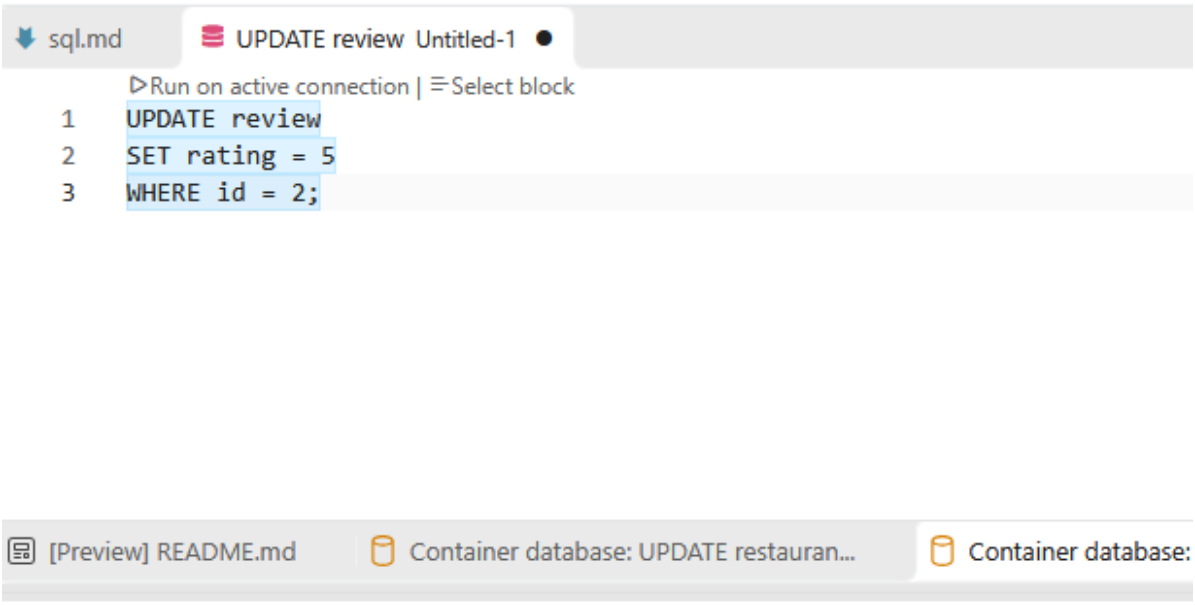


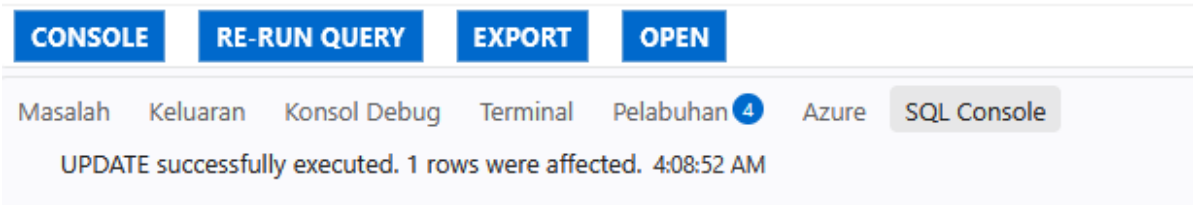




20. Update rating dari salah satu review

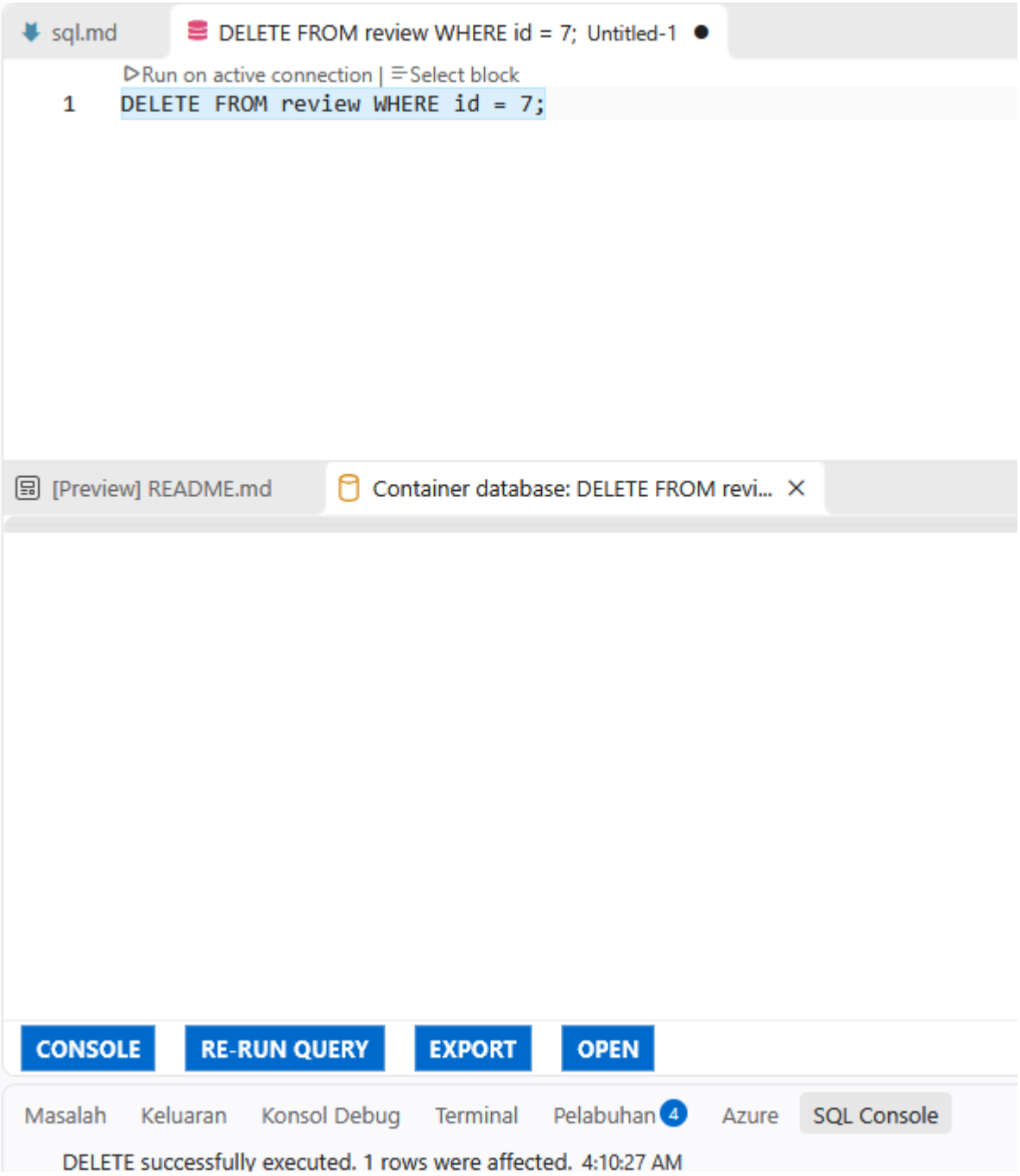
```
UPDATE review SET rating = 5 WHERE id = 2;
```





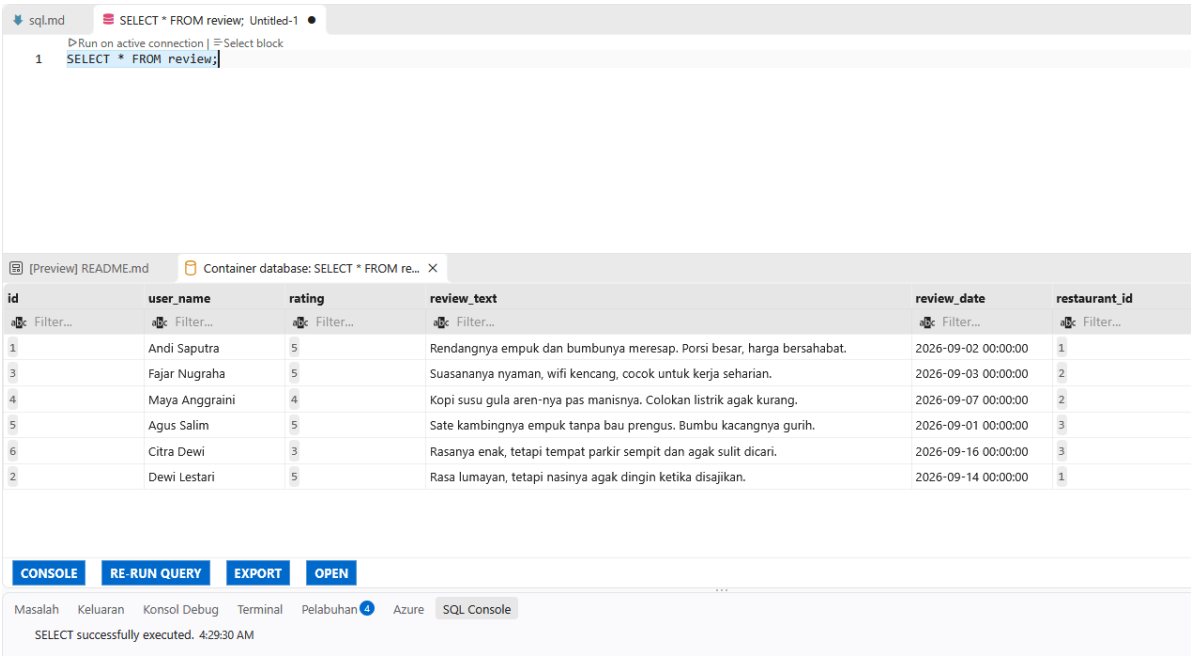
21. Delete review berdasarkan id nya

```
DELETE FROM review WHERE id = 7;
```



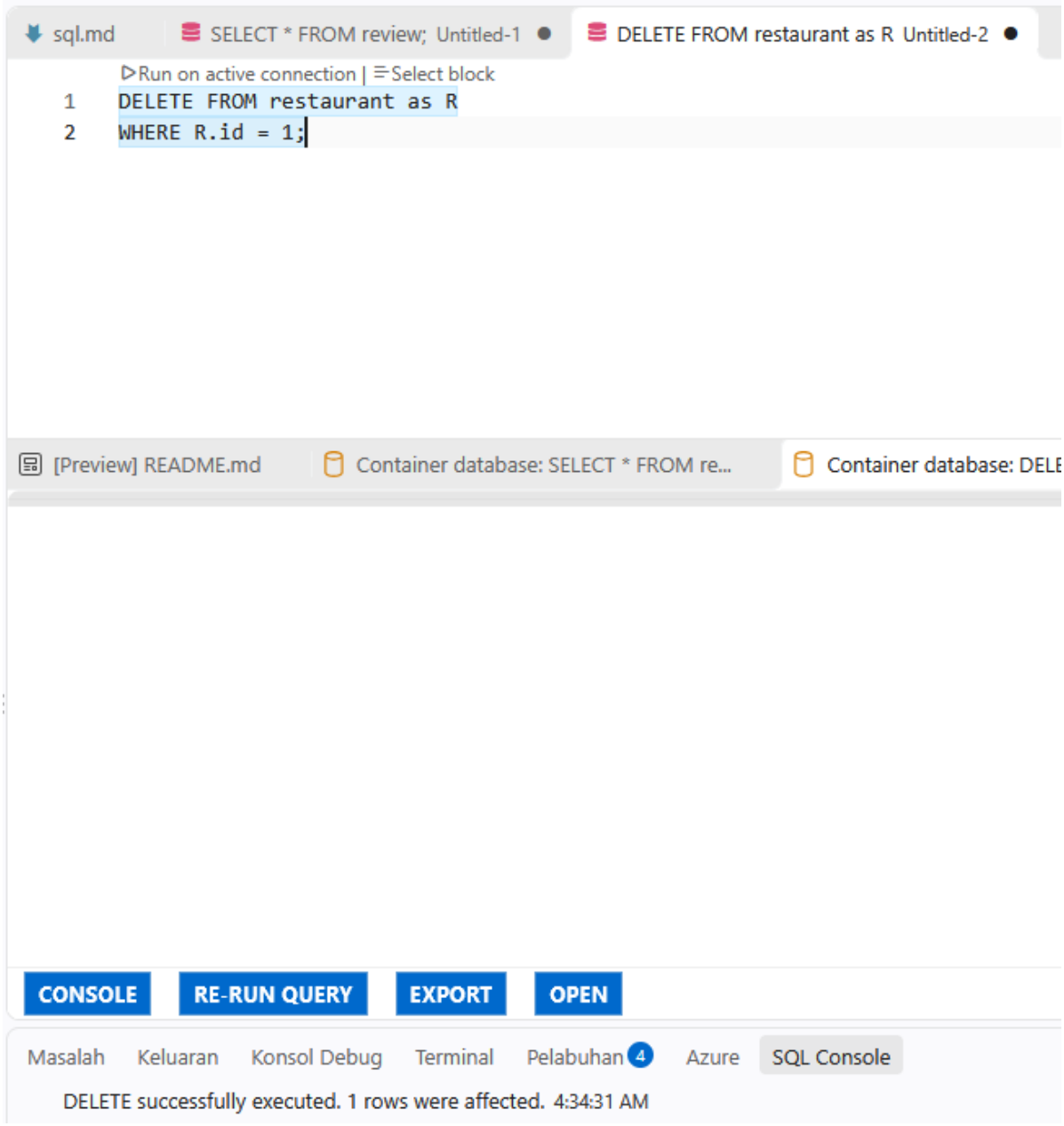
22. Hapus satu data pada tabel restaurant dan pastikan review yang terkait dengannya juga ikut terhapus (menggunakan fitur cascade).

```
SELECT * FROM review;
```



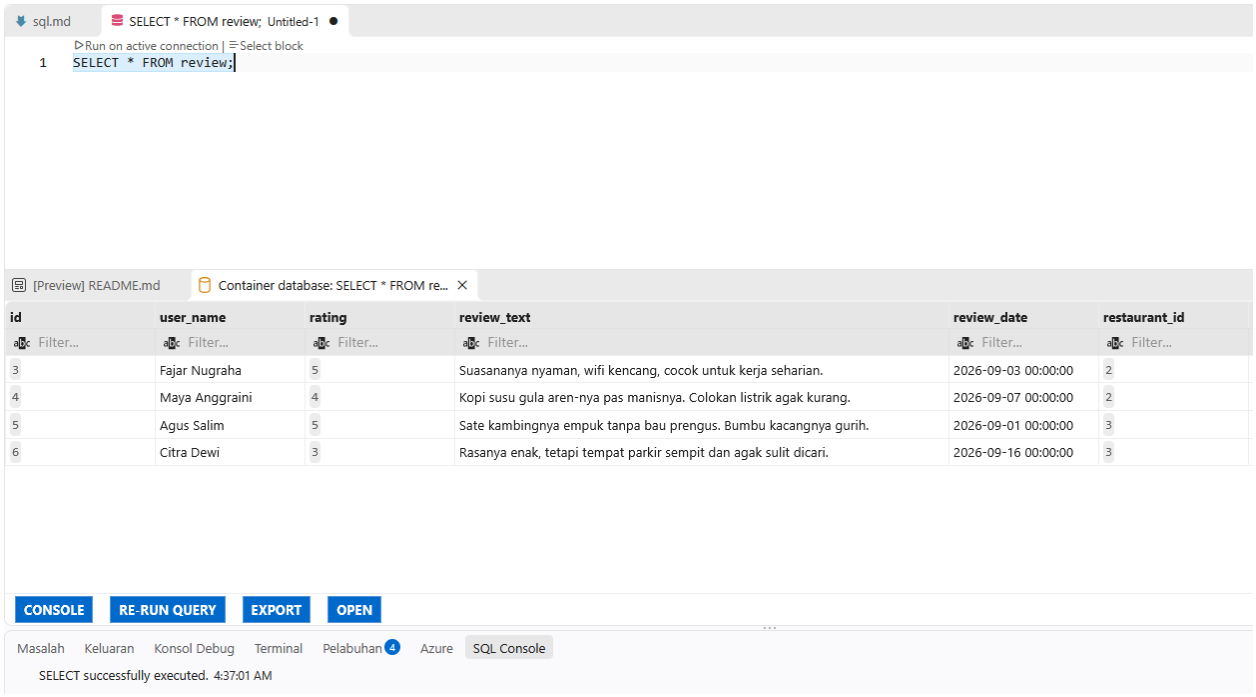
Hapus 1 data restaurant dengan id = 1 dan lihat kembali data review, apakah terhapus dengan fitur cascade?

```
DELETE FROM restaurant as R WHERE R.id = 1;
```


sql.md
SELECT * FROM review; Untitled-1
DELETE FROM restaurant as R Untitled-2
Run on active connection | Select block
1 DELETE FROM restaurant as R
2 WHERE R.id = 1;
[Preview] README.md
Container database: SELECT * FROM re...
Container database: DELE
CONSOLE
RE-RUN QUERY
EXPORT
OPEN
Masalah
Keluaran
Konsol Debug
Terminal
Pelabuhan 4
Azure
SQL Console
DELETE successfully executed. 1 rows were affected. 4:34:31 AM

Sekarang lakukan query ulang untuk tabel review

```
SELECT * FROM review;
```



23. Cari restoran dengan nilai tertinggi berdasarkan rata-rata rating dari seluruh review yang dimilikinya.

```
SELECT r.id,
       r.name,
       ROUND(AVG(v.rating), 2) AS rata_rata_rating
FROM restaurant r
JOIN review v ON v.restaurant_id = r.id
GROUP BY r.id, r.name
ORDER BY AVG(v.rating) DESC
LIMIT 1;
```

sql.md | SELECT r.id, Untitled-1 ●

Run on active connection | Select block

```sql
SELECT r.id,
       r.name,
       ROUND(AVG(v.rating), 2) AS rata_rata_rating
FROM restaurant r
JOIN review v ON v.restaurant_id = r.id
GROUP BY r.id, r.name
ORDER BY AVG(v.rating) DESC
LIMIT 1;
```

[Preview] README.md | Container database: SELECT r.id, ... ×

id	name	rata_rata_rating
Filter...	Filter...	Filter...
2	Kopi Senja	4.50

CONSOLE | RE-RUN QUERY | EXPORT | OPEN

Masalah | Keluaran | Konsol Debug | Terminal | Pelabuhan 4 | Azure | SQL Console

SELECT successfully executed. 5:02:13 AM

24. Cari jumlah review yang telah diterima oleh masing-masing restoran.

```
SELECT r.id,
       r.name,
       COUNT(v.id) AS jumlah_review
FROM restaurant r
LEFT JOIN review v ON v.restaurant_id = r.id
GROUP BY r.id, r.name
ORDER BY jumlah_review DESC, r.id;
```

sql.md | SELECT r.id, Untitled-1 ●

Run on active connection | Select block

```sql
SELECT r.id,
       r.name,
       COUNT(v.id) AS jumlah_review
FROM restaurant r
LEFT JOIN review v ON v.restaurant_id = r.id
GROUP BY r.id, r.name
ORDER BY jumlah_review DESC, r.id;
```

[Preview] README.md | Container database: SELECT r.id, ... ×

id	name	jumlah_review
Filter...	Filter...	Filter...
1	Warung Makan Sederhana	3
2	Kopi Senja	2
3	Sate Pak Haji	2
4	Bakso Mas Joko	0

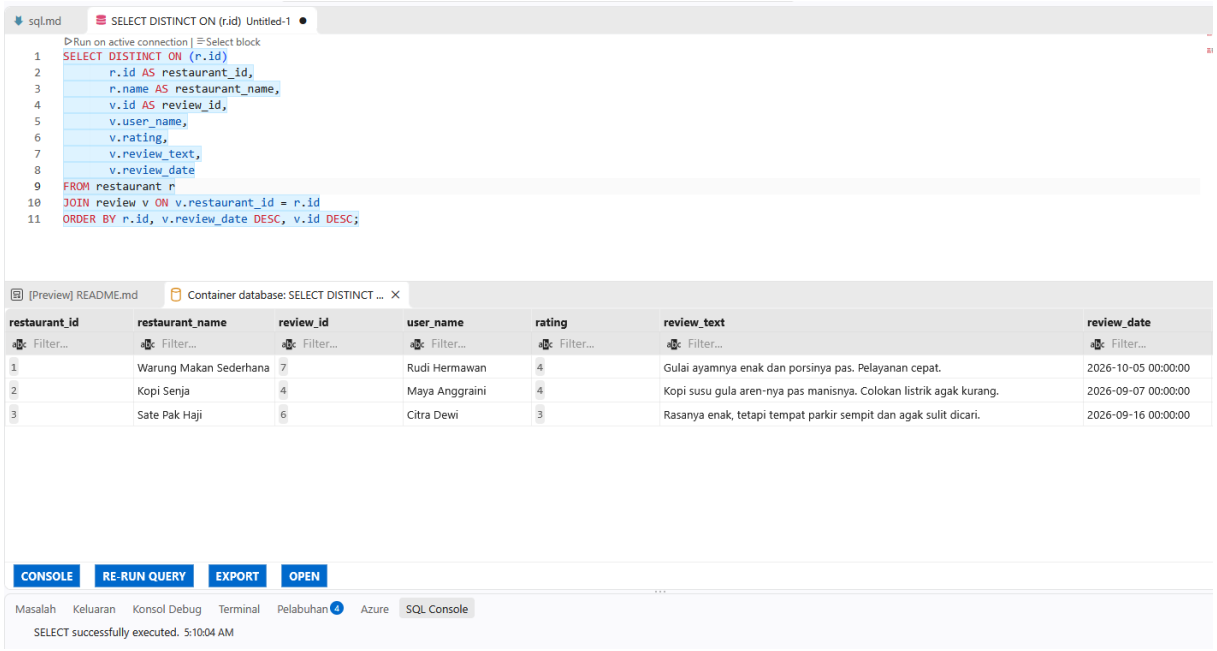
CONSOLE | RE-RUN QUERY | EXPORT | OPEN

Masalah | Keluaran | Konsol Debug | Terminal | Pelabuhan 4 | Azure | SQL Console

SELECT successfully executed. 5:06:09 AM

25. Tampilkan ulasan (review) terbaru untuk masing-masing restoran.

```
SELECT DISTINCT ON (r.id)
        r.id AS restaurant_id,
        r.name AS restaurant_name,
        v.id AS review_id,
        v.user_name,
        v.rating,
        v.review_text,
        v.review_date
FROM restaurant r
JOIN review v ON v.restaurant_id = r.id
ORDER BY r.id, v.review_date DESC, v.id DESC;
```



restaurant_id	restaurant_name	review_id	user_name	rating	review_text	review_date
1	Warung Makan Sederhana	7	Rudi Hermawan	4	Gulai ayamnya enak dan porsinya pas. Pelayanan cepat.	2026-10-05 00:00:00
2	Kopi Senja	4	Maya Anggraini	4	Kopi susu gula aren-nya pas manisnya. Colokan listrik agak kurang.	2026-09-07 00:00:00
3	Sate Pak Haji	6	Citra Dewi	3	Rasanya enak, tetapi tempat parkir sempit dan agak sulit dicari.	2026-09-16 00:00:00

Extra Credit (Optional)

26. Untuk tabel menu sudah ditambahkan pada models.py dan sudah dimigrasikan saat awal build. Lanjut dengan insert 3 data menu untuk tiap tiap restaurant.

```
INSERT INTO menu (restaurant_id, item_name, price, description)
SELECT r.id, m.item_name, m.price, m.description
FROM (VALUES
  -- Warung Makan Sederhana
  ('Warung Makan Sederhana', 'Nasi Rendang',   28000.00, 'Nasi putih dengan rendang daging sapi, sayur daun singkong, dan sambal hijau.'),
  ('Warung Makan Sederhana', 'Gulai Ikan',     26000.00, 'Gulai ikan kakap dengan kuah santan kental dan rempah khas Minang.'),
  ('Warung Makan Sederhana', 'Ayam Pop',       25000.00, 'Ayam kampung rebus goreng sebentar, disajikan dengan sambal tomat.'),

  -- Kopi Senja
  ('Kopi Senja', 'Kopi Susu Gula Aren',        22000.00, 'Espresso, susu segar, dan gula aren dengan es batu.'),
  ('Kopi Senja', 'Caffe Latte',                24000.00, 'Espresso dengan susu steamed, tersedia panas atau dingin.'),
  ('Kopi Senja', 'Croissant Butter',           18000.00, 'Croissant panggang dengan mentega, renyah di luar dan lembut di dalam.'),

  -- Sate Pak Haji
  ('Sate Pak Haji', 'Sate Ayam (10 tusuk)',    25000.00, 'Sate ayam dengan bumbu kacang dan lontong.'),
  ('Sate Pak Haji', 'Sate Kambing (10 tusuk)', 40000.00, 'Sate kambing muda dengan bumbu kecap dan acar bawang.'),
  ('Sate Pak Haji', 'Lontong',                  5000.00, 'Lontong pendamping sate.')
) AS m(restaurant_name, item_name, price, description)
JOIN restaurant r ON r.name = m.restaurant_name;
```

sql.md | INSERT INTO menu (restaurant_id, item_na Untitled-1 ●

Run on active connection | Select block

```sql
INSERT INTO menu (restaurant_id, item_name, price, description)
SELECT r.id, m.item_name, m.price, m.description
FROM (VALUES
    -- Warung Makan Sederhana
    ('Warung Makan Sederhana', 'Nasi Rendang',   28000.00, 'Nasi putih dengan rendang daging sapi, sayur daun singkong, dan sambal hijau.'),
    ('Warung Makan Sederhana', 'Gulai Ikan',     26000.00, 'Gulai ikan kakap dengan kuah santan kental dan rempah khas Minang.'),
    ('Warung Makan Sederhana', 'Ayam Pop',       25000.00, 'Ayam kampung rebus goreng sebentar, disajikan dengan sambal tomat.'),

    -- Kopi Senja
    ('Kopi Senja', 'Kopi Susu Gula Aren',        22000.00, 'Espresso, susu segar, dan gula aren dengan es batu.'),
    ('Kopi Senja', 'Caffe Latte',                24000.00, 'Espresso dengan susu steamed, tersedia panas atau dingin.'),
    ('Kopi Senja', 'Croissant Butter',           18000.00, 'Croissant panggang dengan mentega, renyah di luar dan lembut di dalam.'),

    -- Sate Pak Haji
    ('Sate Pak Haji', 'Sate Ayam (10 tusuk)',    25000.00, 'Sate ayam dengan bumbu kacang dan lontong.'),
    ('Sate Pak Haji', 'Sate Kambing (10 tusuk)', 40000.00, 'Sate kambing muda dengan bumbu kecap dan acar bawang.'),
    ('Sate Pak Haji', 'Lontong',                  5000.00, 'Lontong pendamping sate.')
) AS m(restaurant_name, item_name, price, description)
JOIN restaurant r ON r.name = m.restaurant_name;
```

[Preview] README.md | Container database: INSERT INTO menu... ×

CONSOLE | RE-RUN QUERY | EXPORT | OPEN

Masalah | Keluaran | Konsol Debug | Terminal | Pelabuhan 4 | Azure | SQL Console

INSERT successfully executed. 9 rows were affected. 5:30:11 AM

27. Tulis queri untuk menampilkan setiap restoran beserta menunya dan rata-rata rating dari review nya.

```
SELECT r.id AS restaurant_id,
       r.name AS restaurant_name,
       m.item_name,
       m.price,
       rt.rata_rata_rating
FROM restaurant r
LEFT JOIN menu m ON m.restaurant_id = r.id
LEFT JOIN (
  SELECT restaurant_id,
         ROUND(AVG(rating), 2) AS rata_rata_rating
  FROM review
  GROUP BY restaurant_id
) rt ON rt.restaurant_id = r.id
ORDER BY r.id, m.id;
```

sql.md | SELECT r.id AS restaurant_id, Untitled-1 ●

Run on active connection | Select block

```sql
SELECT r.id AS restaurant_id,
       r.name AS restaurant_name,
       m.item_name,
       m.price,
       rt.rata_rata_rating
FROM restaurant r
LEFT JOIN menu m ON m.restaurant_id = r.id
LEFT JOIN (
    SELECT restaurant_id,
           ROUND(AVG(rating), 2) AS rata_rata_rating
    FROM review
    GROUP BY restaurant_id
) rt ON rt.restaurant_id = r.id
ORDER BY r.id, m.id;
```

[Preview] README.md | Container database: SELECT r.id AS r... ×

restaurant_id	restaurant_name	item_name	price	rata_rata_rating
Filter...	Filter...	Filter...	Filter...	Filter...
1	Warung Makan Sederhana	Ayam Pop	25000.00	4.00
1	Warung Makan Sederhana	Gulai Ikan	26000.00	4.00
1	Warung Makan Sederhana	Nasi Rendang	28000.00	4.00
2	Kopi Senja	Croissant Butter	18000.00	4.50
2	Kopi Senja	Caffe Latte	24000.00	4.50
2	Kopi Senja	Kopi Susu Gula Aren	22000.00	4.50
3	Sate Pak Haji	Lontong	5000.00	4.00
3	Sate Pak Haji	Sate Kambing (10 tus...	40000.00	4.00
3	Sate Pak Haji	Sate Ayam (10 tusuk)	25000.00	4.00
4	Bakso Mas Joko	NULL	NULL	NULL

CONSOLE | RE-RUN QUERY | EXPORT | OPEN

Masalah | Keluaran | Konsol Debug | Terminal | Pelabuhan 4 | Azure | SQL Console

SELECT successfully executed. 5:33:27 AM